

Committed acceptance objects

D1: runtime graph

$$G_t = (\mathcal{V}_t, E_t, \tau_V, \tau_E, \text{attr}_t)$$

D2: replayable certificate

$$Z = (c, S, \Pi, \Gamma, \mathcal{A}, \chi, \text{meta})$$

D3: grounded acceptance

$$\text{Check}(Z; G_t) = V_H V_\Pi V_\Gamma V_-$$

Checker-relative acceptance is not a world-truth proof.

Structure and executable contract

D4: (δ, κ, ζ) -separated support paths.

Replay intervention $G_t^{\setminus e}$; responsibility $\text{Resp}(e; Z, G_t)$.

Algorithm 1: proof-carrying agentic verification.

Branches $\rightarrow$ individual checks
 $\rightarrow$ separated certificates $\rightarrow$ risk-aware action.

Operational separation is not causal independence; the algorithm is not a proof node.

Declared assumptions

A1: audit channels recomputable; uncovered in-scope taxonomy mass is zero or explicitly charged as ε_{tax} .

A2: bounded, conditionally i.i.d. replay effects; bounded calibration error; Lipschitz secondary controller.

Outside guarantee: source-to-world residual ε_{src} .

Minimal acceptance and finite audit

Theorem C.1: each conjunct has a separating witness.

Lemma C.1: represented failures plus ε_{tax} .

Proposition C.1: finite-sample envelope

$$\Pr(\text{ContractBad}) \leq B_{\text{cov}} + \pi_{\text{unc}} + \varepsilon_{\text{tax}}^{\text{cov}}.$$

Theorem C.2: width $\Omega(\sqrt{\log(1/\delta)/n})$ and unavoidable uncovered-mass charge.

Dependence and redundancy

Theorem C.3: all-fail likelihood-ratio / Rényi envelope
 $\Pr(\bigcap_{i \in I} E_i) \leq \rho_I^{|I|-1} \prod_{i \in I} p_i.$

Theorem C.4:

$$q_{\text{cm}} \leq \Pr(\bigcap_i E_i).$$

Redundancy cannot certify below the common-mode floor.

ρ -gate: OPEN / CLOSED / INSUFFICIENT-EVIDENCE.

Certified operating-point consequence:

$$B_{\text{audit}} + \bar{\rho}^{k-1} \bar{\epsilon}^k \leq \alpha_*,$$

Acceptance event $A_t^{(k,q)}(c).$

Shift, diagnosis and selection

Theorem C.5: restricted-TV transfer for $E \in \mathfrak{F}_Z$; the domain alarm is not an upper risk certificate.

Proposition C.2: responsibility concentration; controller regret $2L_{\text{ctrl}}\epsilon_{\text{cal}}$ under its declared model.

Lemma C.2: exact $\Delta = S + V$ identity.

An identity does not identify an empirical mechanism. Logged replay is not real-world causal attribution.

Witness and audit stress

Figure 16: force one conjunct to fail while others pass.

Figure 15: probe count, uncovered mass, coverage and audit envelope.

Figure 4: declared theory geometry.

Dependence and replay stress

Figure 9: provenance, tools, evidence and semantic overlap.

Figure 17: co-failure and common-mode stress.

Figure 8: fresh drift tests brittleness; it does not retroactively invalidate a committed snapshot.

Transfer, diagnosis and selection stress

Figure 6: certificate-measurable shift.

Figure 18: ranked responsibility and unresolved cases.

Figure 5: coverage and risk changes, without unsupported S/V attribution.

Controller sensitivity: calibration error is separate from broader cost/harm sensitivity.

Solid: formal dependency and test design. Dashed: empirical stress.

Stress tests challenge applicability; they do not prove assumptions.